

IT3060– Human Computer Interaction

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Lab Exercise 04

Campus: Malabe Campus

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Activity 1: Define Usability Goals

Usability objectives were developed based on the issues mentioned in the case study (lower retention rate, decrease in app-store rating from 4.6 to 3.8 stars), as well as on three major pain points that the redesign would solve.

1.1 Measurable Usability Objectives

Goal	Metric	Target	Business Alignment
Overall usability	System Usability Scale (SUS) score	≥ 80 (baseline round: ≥ 70)	Reverses the drop in rating from 4.6 to 3.8 stars; SUS 80+ indicates an 'excellent' design according to the industry standard.
Task effectiveness	Task completion rate	$\geq 90\%$ across core tasks	Directly targets the 68% onboarding drop-off reported in the case study.
Task efficiency	Average task completion time	$\leq 60s$ for logging tasks; $\leq 90s$ for setup tasks	Solves the challenge of having to record workouts and nutrition.
Error tolerance	Errors per task	≤ 1 minor error; 0 critical errors	Facilitates confidence in the ease of use of the dashboard.
Perceived ease	Single Ease Question (SEQ), 1–7 scale	Mean ≥ 5.5	Captures subjective friction not visible in completion metrics alone.

1.2 Priority Goals by Pain Point

Pain Point	Prioritized Usability Goal
Navigation	Users are able to navigate through all five essential sections (Home, Workout, Progress, Community, Nutrition) in ≤ 2 clicks, with $\geq 95\%$ of users finding the proper section within their first attempt.
Dashboard usability	$\geq 90\%$ of respondents can accurately interpret and follow the AI Daily Flow suggestion within 15 seconds of visiting the Home Dashboard page..
Progress tracking	$\geq 90\%$ of participants are capable of understanding their graph that shows progress every week and tell whether they are on or off track.

Activity 2: Prepare Test Tasks and Success Criteria

#	Surface	Task Scenario	Success Criteria
T1	Prototype	"You have just launched the FitFlow app for today. Locate and begin today's workout recommended by AI."	Success: workout started within 30s, 0 critical errors.
T2	Existing app	"Log your having done 3 sets of squats just now."	Success within 45s; note number of taps required.
T3	Prototype	"Log your lunch using the camera."	Success within 30s; camera flow reached without help.
T4	Existing app	"Log your lunch through the typing option."	Success within 90s; used for friction comparison against T3.
T5	Prototype	"Review how you performed in the workouts this week."	Correctly reads chart and states on/off-track status within 20s.
T6	Prototype	"Find a community circle you can join and give motivation to one of the members."	Circle located and 'Cheer' sent within 60s.
T7	Prototype	"You failed to perform yesterday's workout. Get your streak back."	Recovery action found and completed within 45s, 0 critical errors.
T8	Prototype	"Navigate to the Nutrition Logger starting from the Home Dashboard and return to Home."	Completed in ≤ 2 taps each way; used as the core navigation efficiency check.

2.1 Task Script (Moderator Read-Aloud)

- "I am going to present you with a brief scenario and ask you to attempt to finish it with the help of the app. You cannot do anything wrong here as we want to test the app and not you."
- "You will need to talk aloud about everything that you see and feel."
- "Don't hurry. Let me know when you think that you have finished the task or want to quit."

2.2 Post-Task Survey Questions

- Single Ease Question (SEQ): "How easy or difficult was this task, overall?" (1 = Very difficult; 7 = Very easy)
- "What, if anything, made this task more difficult than necessary?" (Open-ended text response)

- "Were there any parts of this screen that were confusing or unexpected?" (Open-ended text response)

Activity 3: Recruit Participants

3.1 Screening Criteria

- Age 18–55 years, mixed between bands to represent FitFlow’s user population.
- Fitness level: mix of beginners, intermediates, and advanced (based on the three personas in Lab Exercise 02/03).
- User type: current active users, lapsed users (60+ days inactive), and new users unaware of FitFlow.
- Device platform: iOS and Android users, mixed, in order to address platform-specific problems.
- Familiarity with technology: mix, in order not to over-sample power users.

3.2 Recruitment Channels

- In-app invite for current active users.
- Re-invite of previous interview and diary-study participants (lapsed users, who have agreed to be contacted in the future).
- Recruiting panel of new users unaware of FitFlow.

3.3 Participant Roster (10 participants)

ID	Segment	Age	Fitness Level	Device	Status
U01	Busy Professional	34	Intermediate	iOS	Current user
U02	New Parent	29	Beginner	Android	Current user
U03	Fitness Enthusiast	41	Advanced	iOS	Current user
U04	Busy Professional	38	Intermediate	Android	Lapsed user
U05	New Parent	31	Beginner	iOS	Lapsed user
U06	General	26	Beginner	Android	New user
U07	General	45	Intermediate	iOS	New user
U08	Fitness Enthusiast	37	Advanced	Android	Current user
U09	Busy Professional	50	Intermediate	iOS	Lapsed user
U10	New Parent	33	Beginner	Android	New user

3.4 Screening Questionnaire (Summary)

- Age range; occupation; typical number of times one exercises weekly.
- Use of FitFlow or similar fitness applications currently or in the past.
- Type of device used and familiarity with mobile applications (scale 1-5).
- Availability during the proposed testing period; permission for recording.

3.5 Scheduling and Incentives

- 45-minute moderated remote sessions conducted over video call, one participant at a time, across a 5-day testing window.
- Each participant receives a \$25 gift voucher as a thank-you incentive.

Activity 4: Conduct Think-Aloud Testing

4.1 Session Protocol

- Greeting and informed consent (recording, data collection, freedom to discontinue participation at any moment).
- Moderating questions: current exercise behavior, experience using fitness applications.
- Narration instruction and a warm-up task not related to FitFlow to get accustomed to speaking one's mind.
- Task completion: participant performs T1-T8 consecutively, the moderator remains silent except when asking neutral questions.
- SEQ after each task; SUS survey and a short debriefing interview after the session.

4.2 Neutral Probing Questions (used sparingly, only when a participant goes silent)

- "What are you thinking right now?"
- "What did you expect to happen when you tapped that?"
- "What would you do next?"

4.3 Recording and Observation

Each session was screen- and audio-recorded with consent. A note-taker logged timestamps for task start/end, errors, and notable verbal reactions in a shared observation sheet, separate from the moderator, to reduce bias in real-time note-taking.

Activity 5: Measure Task Success and Completion Time

5.1 Task-Level Metrics (aggregated across 10 participants)

#	Task	Success Rate	Avg. Time	Avg. Errors	Avg. SEQ
T1	Start AI-recommended workout	90%	18s	0.2	6.4
T2	Log squats (existing app)	80%	52s	1.1	4.2
T3	Log lunch via camera (prototype)	90%	22s	0.3	6.6
T4	Log lunch manually (existing app)	70%	78s	1.6	3.6
T5	Check weekly progress	90%	16s	0.1	6.5
T6	Find circle + send encouragement	80%	48s	0.6	5.8
T7	Recover a missed streak	80%	35s	0.4	6.0
T8	Home → Nutrition → Home navigation	100%	9s	0.0	6.8

5.2 SUS Score

The standard 10 - item SUS questionnaire was administered after each session.

Participant	SUS Score
U01	78
U02	68
U03	82
U04	70
U05	66
U06	74
U07	72
U08	84
U09	69
U10	71
Mean (Round 1)	73.4

Activity 6: Record Usability Issues and Severity Ratings

Problems were identified using observational notes, recording sessions, and open-ended responses from surveys and categorized based on their severity as Critical (stops users from completing the task), Major (creates serious problems/difficulties), Minor (creates pauses but is reversible), and Enhancement (nice to have but not essential).

ID	Issue Description	Task / Screen	Frequency	Severity	Example Reaction (paraphrased)
I01	Manual nutrition entry in the existing app requires too many fields before saving.	T4 / Nutrition (existing)	7 / 10	Major	Participant said typing everything felt like more effort than the meal was worth.
I02	Some participants tapped the streak number expecting it to show a recovery option, rather than using the intended button.	T7 / Home Dashboard	4 / 10	Minor	One participant assumed the number itself was tappable.
I03	The 'Cheer' action label was unclear to two new users unfamiliar with fitness-app conventions.	T6 / Community Feed	2 / 10	Minor	A participant asked what tapping the heart-like icon would actually do.
I04	Camera-based food logging occasionally required a retake due to unclear framing guidance.	T3 / Nutrition Logger	3 / 10	Minor	Reported needing a second attempt to get a usable photo.
I05	Existing app's squat-logging flow required navigating through two extra confirmation screens.	T2 / Workout (existing)	6 / 10	Major	Participant commented it took longer to log the set than to do it.
I06	One participant could not locate the weekly chart without scrolling further than expected.	T5 / Progress Tracking	1 / 10	Enhancement	Minor hesitation before scrolling down.
I07	No visible confirmation animation after starting the AI workout, leaving momentary uncertainty it had registered.	T1 / Home Dashboard	3 / 10	Minor	Participant paused, unsure whether the tap had worked.

Activity 7: Prepare Improvement Recommendations

7.1 Prioritized Recommendations

Issue	Recommendation	Priority	Expected Impact
I01 / I05	Discontinue the current multi-step logging process and opt for the improved camera nutrition logger and one-click exercise logging that has been tested in T1-T3.	High (Quick win — already built in prototype)	Directly resolves the most frequent and severe issues (7/10 and 6/10 participants affected).
I07	Provide an animation or message that confirms the initiation of a workout through the AI Daily Flow card.	High (Quick win)	Removes uncertainty at a high-traffic entry point with minimal design effort.
I02	Make the tappable number the actual streak number, like the recover button, as users will be expecting that.	Medium	Reduces hesitation for a moderately common misclick pattern.
I04	Add an on-screen framing guide or auto-capture cue to the camera nutrition logger.	Medium	Reduces retake rate and reinforces trust in the camera-logging feature.
I03	Change or add a short name for the "Cheer" function (such as "Cheer  "), along with a one-time tooltip for first-time users.	Medium	Improves clarity for new users without disrupting existing user familiarity.
I06	Raise the weekly chart to be closer to the top of the Progress page for visibility at standard device sizes.	Low	Minor layout adjustment; affected only 1/10 participants.

7.2 Final Usability Test Report — Summary

The results of Round 1 ($n = 10$) showed a mean SUS score of 73.4, with the redesigned prototype flows (T1, T3, T5, T6, T7, T8) outperforming the corresponding existing-app flows (T2, T4) in terms of completion, speed, and SEQ, which confirms in quantitative form the hypothesis about the redesign fixing the problems of personalization, tracking friction, and progress visualization found in the prior research. Two Major issues (I01, I05) were fully solved by the current features in the prototype without requiring any further design, which means these are quick wins. The remaining Minor issues are minor adjustments of low risk, good to implement in the next design iteration. Following the pattern shown in the case study of SUS increasing from 68 to 87 after three iterations, the second round using Activity 7 recommendations will take the mean SUS score from 73.4 to above 80, the goal set up in Activity 1.